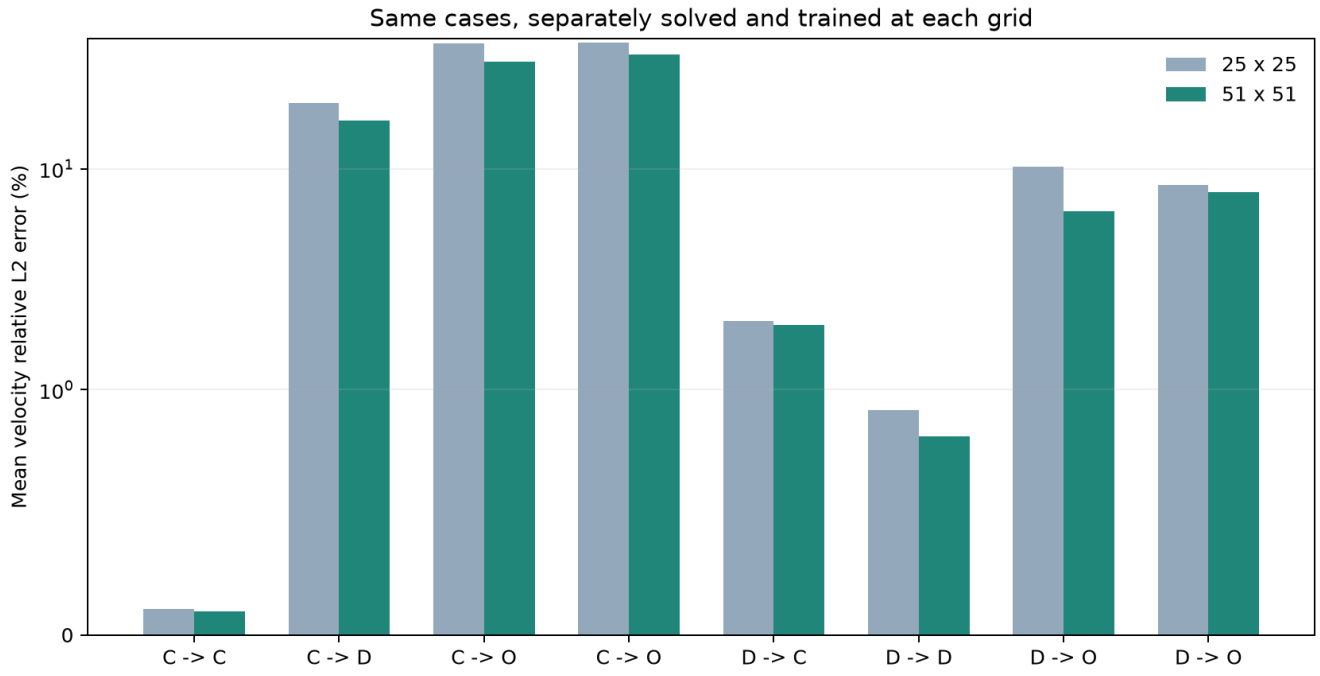
51 x 51 numerical refinement

Matched Stokes fields and Navier-Stokes labels were solved again on 51 x 51 nodes.

The retained case manifest is unchanged; tests are regression cases, not fresh blind data.

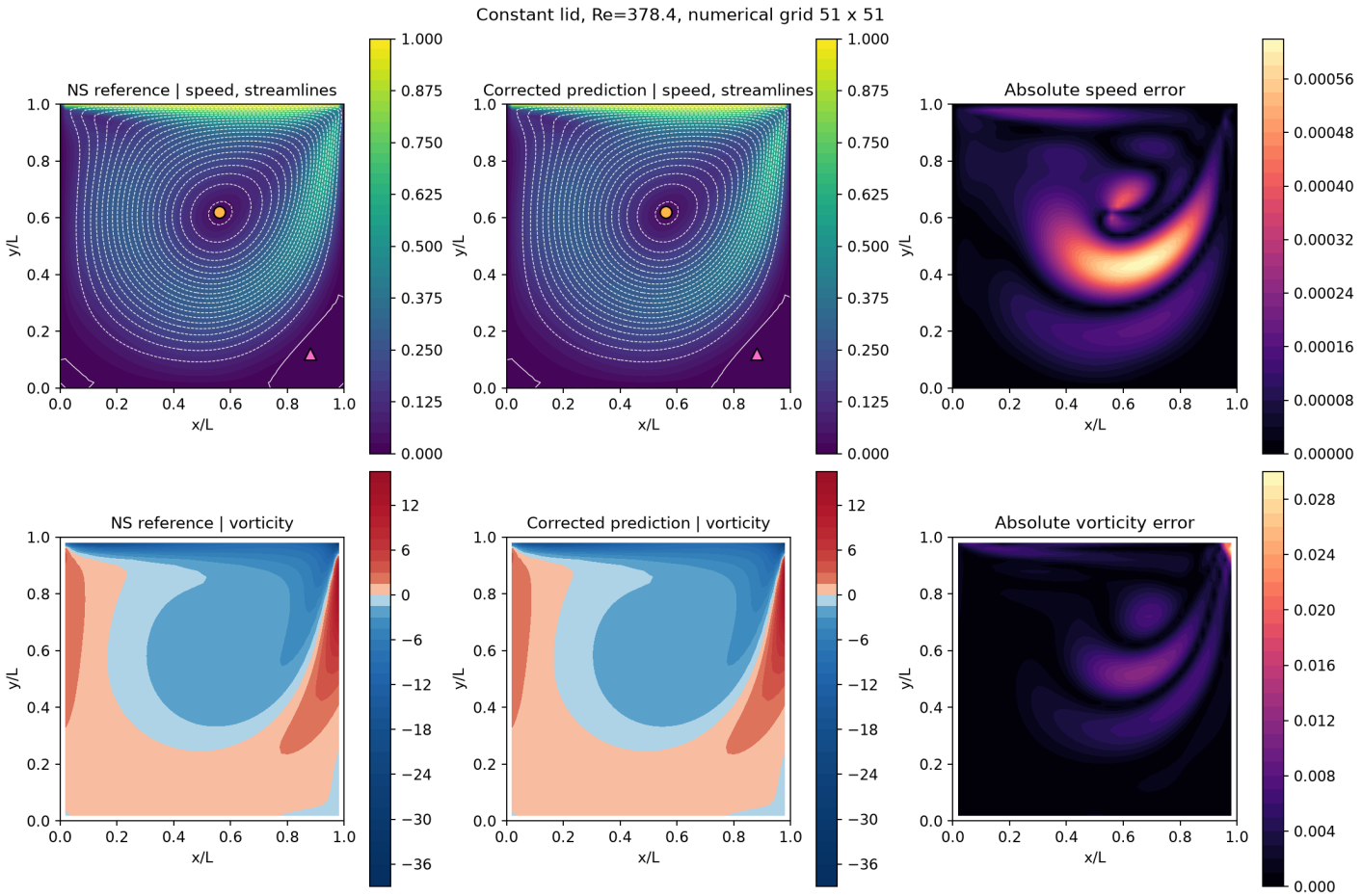
The selected POD-MLP architecture was retrained from scratch at 51 x 51.



Train -> test	**25 x 25**	**51 x 51**
constant -> constant	0.107%	0.097%
constant -> diverse	19.920%	16.555%
constant -> ood_re	36.982%	30.700%
constant -> ood_shape	37.192%	33.068%
diverse -> constant	2.044%	1.960%
diverse -> diverse	0.917%	0.810%
diverse -> ood_re	10.204%	6.425%
diverse -> ood_shape	8.428%	7.878%

Errors compare models with separate numerical labels on each grid.

Constant lid: speed, streamlines, vorticity



Mean CFD 25-to-51 velocity change at coarse nodes: 20.59%

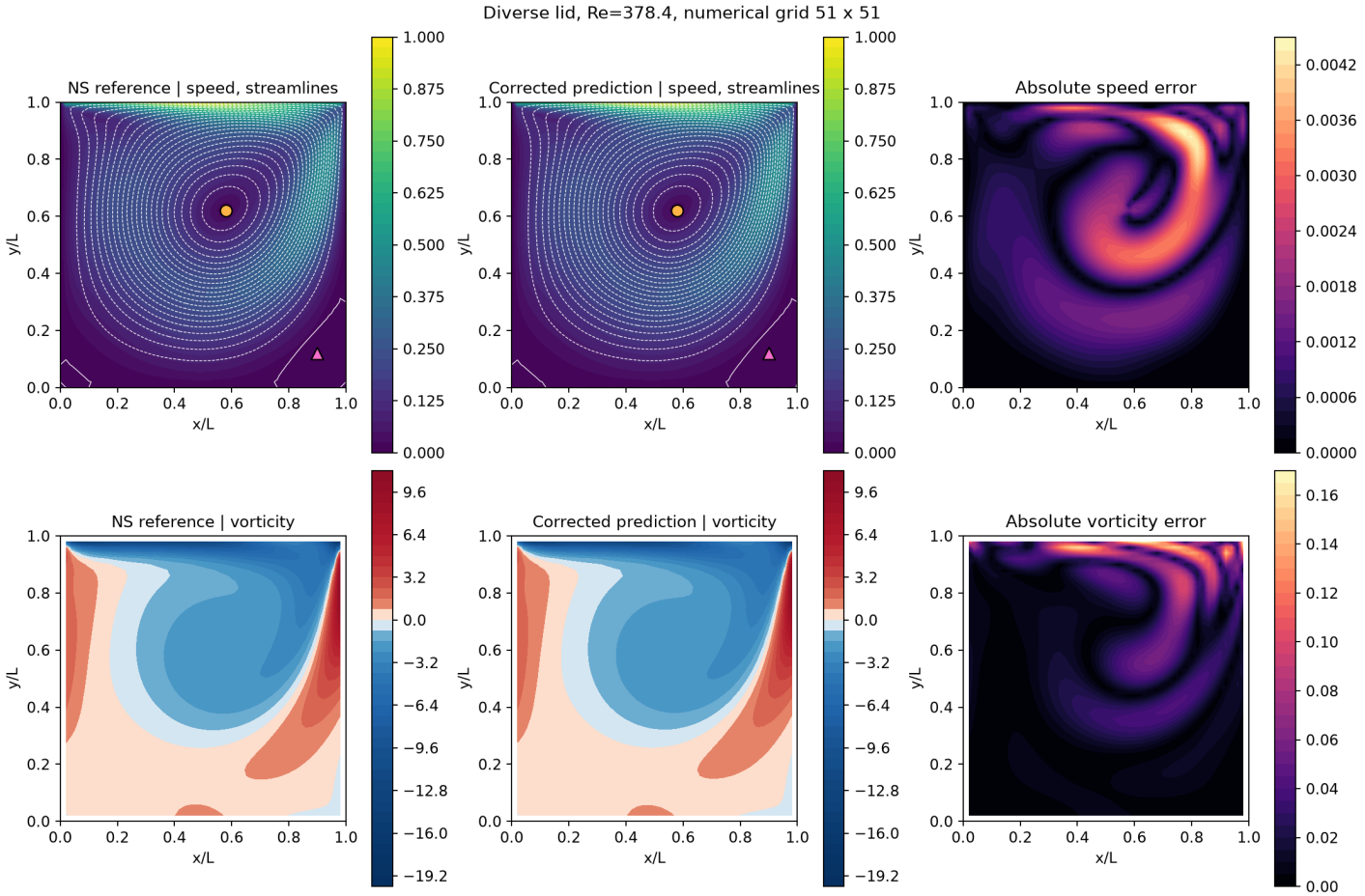
Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.

Inter-grid differences are not estimates of either grid's true error without further assumptions.

Small same-grid surrogate error establishes label fidelity, not independent physical accuracy.

Diverse lid: speed, streamlines, vorticity



Mean CFD 25-to-51 velocity change at coarse nodes: 17.81%

Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.

Inter-grid differences are not estimates of either grid's true error without further assumptions.

Small same-grid surrogate error establishes label fidelity, not independent physical accuracy.